Speech Analysis

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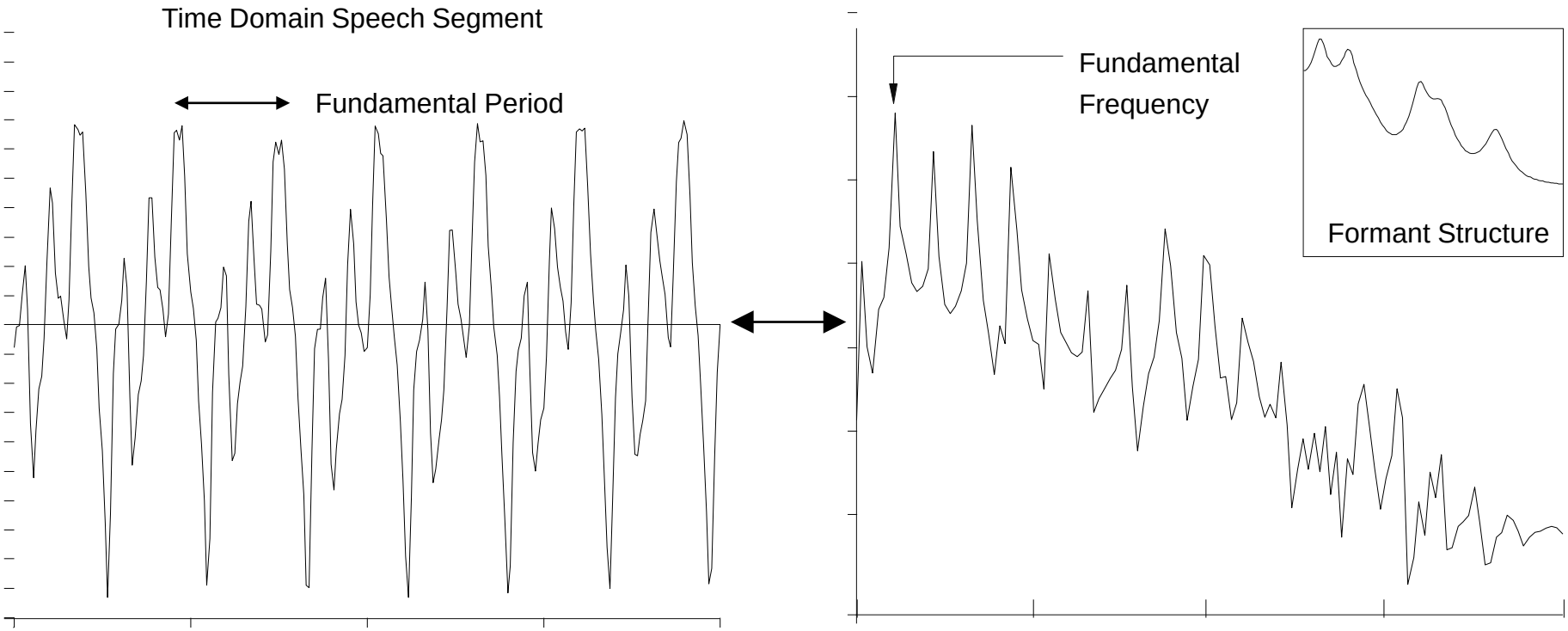
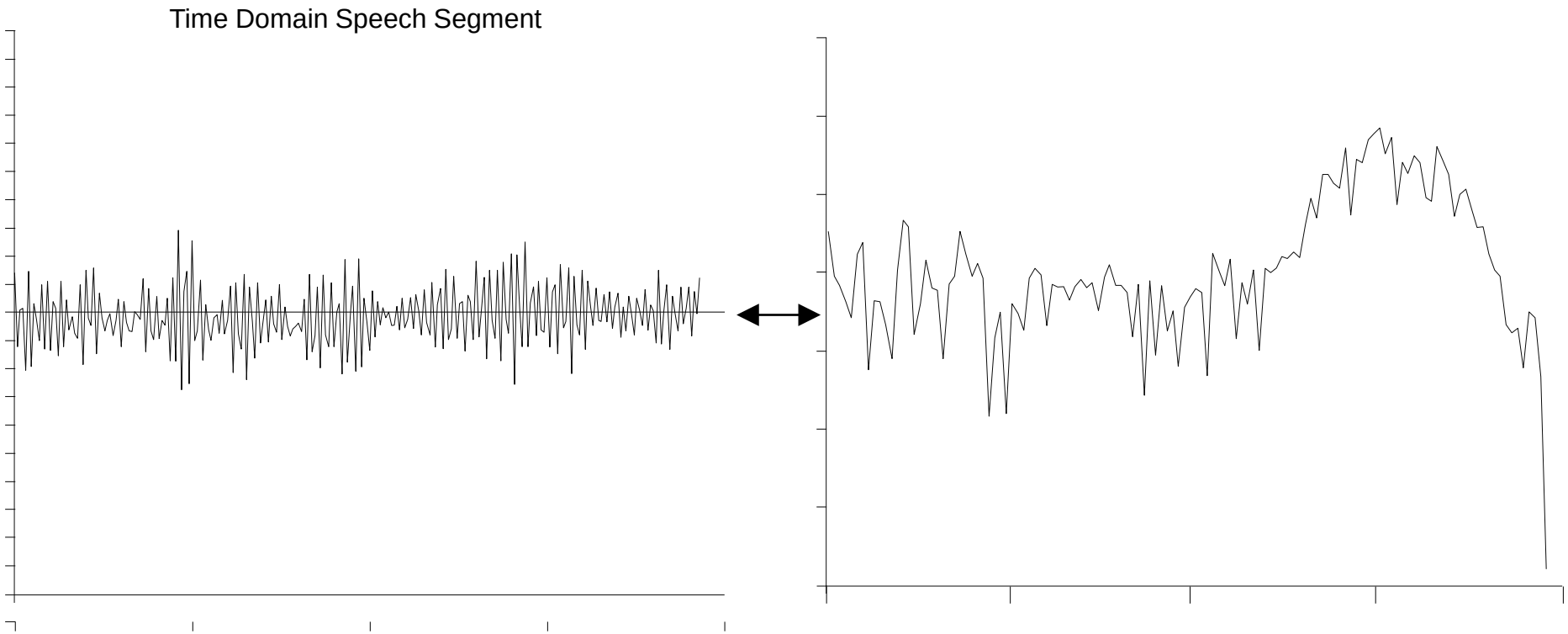
Speech Signals

- **Non-stationary, i.e., change over time**
- **Quasi-stationary over short segments, typically 5-20 ms**
- **Classified as**
  - voiced (e.g., /a/, /i/, etc),
  - unvoiced (e.g.,/sh/),
  - or mixed

Example: Voiced Speech Signal and its Short Time Spectrum

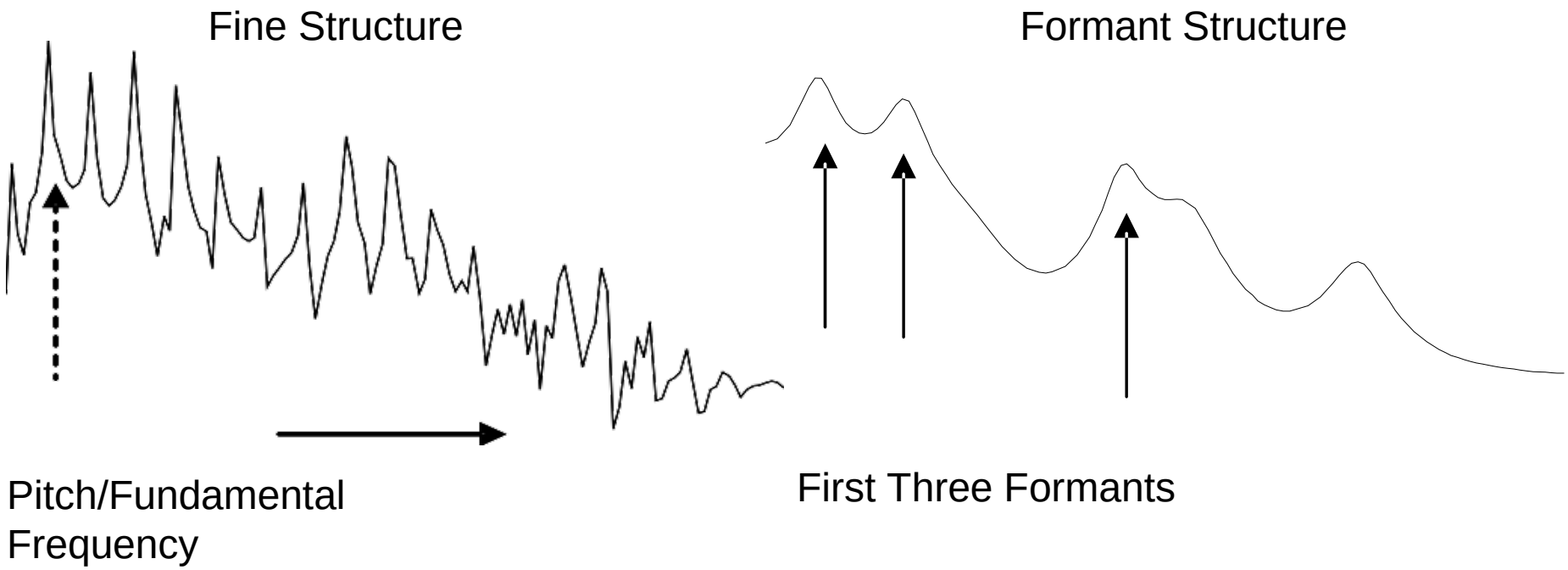


Example: Unvoiced Speech Signal and its Short Time Spectrum



Fine and Formant Structures

The short-time spectrum of voiced speech is characterized by its fine and formant structures:



Fine Structure

Fine structure: the rapid oscillations close to the fundamental frequency

- Harmonic structure is a consequence of the quasi-periodicity of speech and may be attributed to the vibrating vocal cords.
- The first peak of the fine structure (other than the DC component) is the **fundamental or pitch frequency** and is usually higher for female speakers relative to male speakers.

Formant Structure

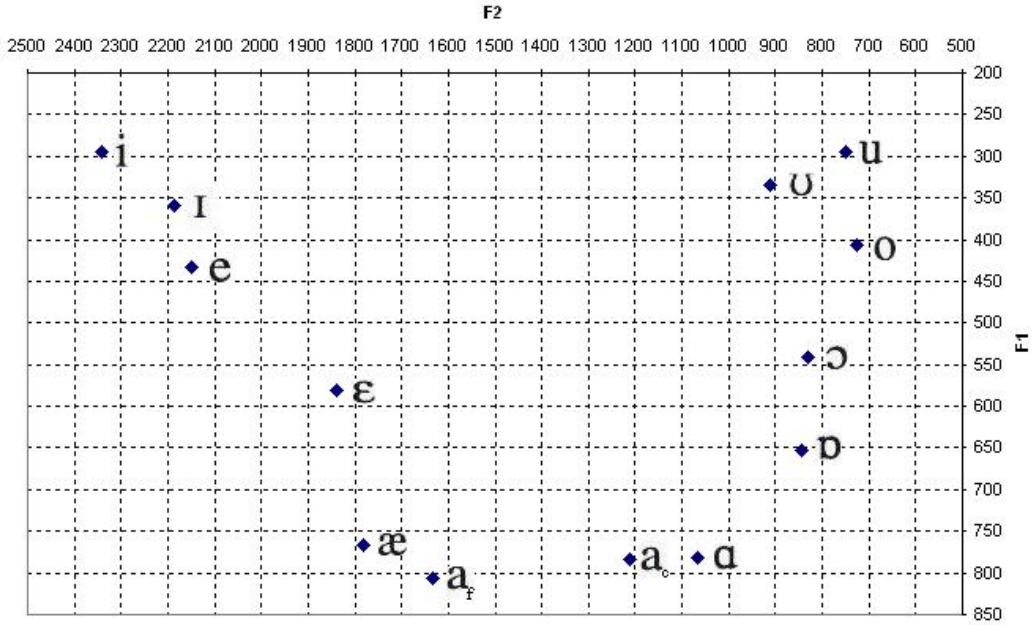
Formant structure: slower amplitude modulations superimposed on the fine structure

- A result of the interaction of the source and the vocal tract.
- The spectral envelope is characterized by a set of peaks which are called **formants**.
- The formant frequencies characterize the intelligibility of speech and are often parameterized and used for recognizing and synthesizing phonemes.

Formant Frequencies of Vowels

- **Formant frequencies vary from person to person due to gender, accent, etc.**

- **Example: https://linguistics.ucla.edu/people/hayes/103/Charts/VChart/**



Question

Which of the following statements is correct?

a) Speech signal is non-stationary over very short segments.

b) Fine structure is the slower amplitude modulations superimposed on the formant structure.

c) Formant frequencies of speaking the same vowel are the same for everyone.

d) The first peak of the fine structure is the pitch frequency.

Autocorrelation Function

- $x[n]$ **is a real valued signal**

- **Autocorrelation:**
  - $R_x[n] = \sum_{k=-\infty}^{\infty} x[k]x[k-n], -\infty \leq n \leq \infty$
  - $R_x[n] = x[n] * x[-n]$ (convolution)

- **Given** $x[n] \overset{DTFT}{\longleftrightarrow} X\left[e^{j\omega}\right]$
  - Time reversal property: $x[-n] \overset{DTFT}{\longleftrightarrow} X\left[e^{-j\omega}\right]$
  - Conjugate symmetry for real signals: $X\left[e^{j\omega}\right] = X^*\left[e^{-j\omega}\right] \Rightarrow X\left[e^{-j\omega}\right] = X^*\left[e^{j\omega}\right]$
  - Convolution property: $x[n] * y[n] \overset{DTFT}{\longleftrightarrow} X\left[e^{j\omega}\right]Y\left[e^{j\omega}\right]$

- **We get**
  - $R_x[n] \overset{DTFT}{\longleftrightarrow} X\left[e^{j\omega}\right]X^*\left[e^{j\omega}\right] = \left|X\left[e^{j\omega}\right]\right|^2$

Autocorrelation Function

Properties

- $R_x[n]$ is a real, even function, i.e., $R_x[n] = R_x[-n]$
- $|R_x[n]| \leq R_x[0]$ for all *n*, i.e., maximum value at *n = 0.*

Used to identify repeated patterns within a signal

Applications

- Speech recognition
- Audio and video processing
- Radar and sonar systems
- Time series analysis

How to estimate pitch frequency?

- Pitch/fundamental frequency for voiced speech segment can be extracted from the autocorrelation function.
- The peak of the autocorrelation function will indicate the pitch frequency.
- The autocorrelation function can be computed as the inverse Fourier transform of the signal spectrum.

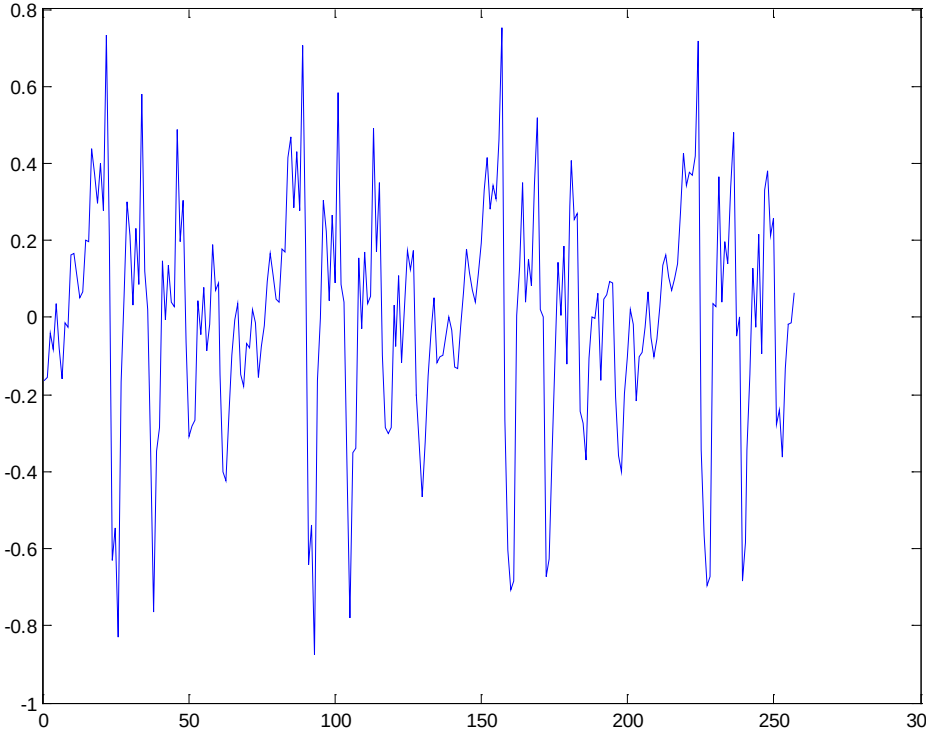
Pitch Estimation Algorithm

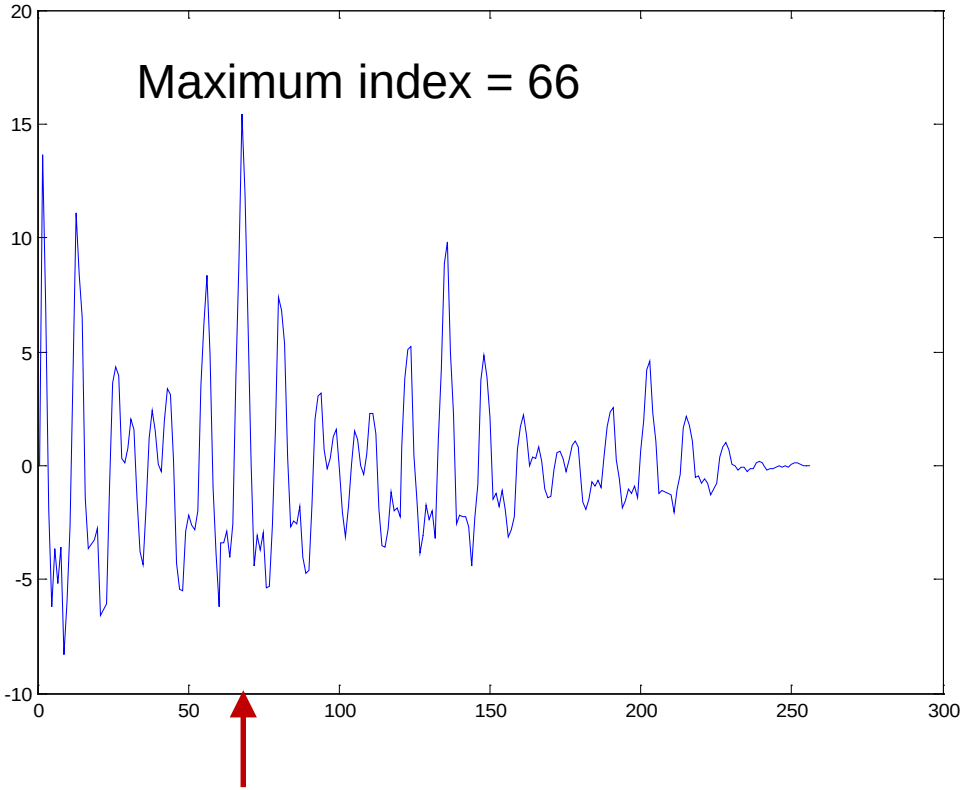
- **Compute FFT on the voiced segment *x[n].***
- **Compute the signal spectrum (magnitude squared) from the complex FFT output.**
- **Compute IFFT (inverse FFT) of the signal spectrum to obtain the autocorrelation function.**
- **Search for the maximum peak value of the autocorrelation function (excluding DC component, which is self correlation).**
- **Convert the index of the maximum value to the estimated pitch frequency.**

Example

Period = 66 samples =>

For a sampling frequency of 8000Hz, the pitch/fundamental frequency is 1/T = 1/(66*Ts) = fs/66 = 121Hz





Time domain voice samples

Autocorrelation function

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